

Energy-Optimal Hop Count in Equal-Hop Multi-Hop Wireless Links: An Analytical and Computational Baseline Model

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Abstract—While working on a multi-hop mesh communication system, a practical question came up: if a source and destination are separated by a fixed distance, how many hops should be used to transfer a packet efficiently? A direct link avoids the energy cost of additional relay nodes, but requires higher transmit power. Adding relay nodes shortens the individual links and reduces the required transmit power, while also introducing additional transmitter and receiver energy at every hop. This work develops a simple analytical and computational model to study this trade-off under ideal free-space conditions. Friis transmission loss is combined with receiver sensitivity, power-amplifier efficiency, transmitter and receiver circuit power, packet duration, and a maximum transmit-power constraint. For equal-length hops, the resulting packet-energy objective takes the form $E(H) = A/H + BH$, giving a continuous stationary point and a constrained integer optimization problem. A reproducible V1 experiment suite was used to examine the effects of distance, circuit power, carrier frequency, PA efficiency, receiver sensitivity, packet size, data rate, and maximum transmit power. For the 500-m baseline, the integer optimum is one hop, requiring approximately 4.031 dBm (2.53 mW) of RF transmit power and 21.78 μ J per packet. The model is intentionally idealized and is intended to provide a transparent baseline for later models incorporating environmental and channel-dependent effects.

Index Terms—multi-hop communication, energy optimization, wireless networks, hop count, Friis transmission, low-power networking, analytical modeling

I. INTRODUCTION

Multi-hop communication replaces one long wireless link with a sequence of shorter links. Shorter links can reduce the RF transmit power required to reach the receiver. However, every additional hop also requires another node to receive and transmit the packet, which adds energy consumption. Therefore, increasing the number of hops does not necessarily minimize the total energy required to deliver a packet.

The starting point for this work was a practical problem encountered while working on a multi-hop mesh communication system. For a fixed source-to-destination distance, there can be several possible ways to divide the communication path into hops. A direct connection uses the fewest nodes but may require substantially more transmit power, while additional relay nodes reduce the distance of each individual transmission at the cost of additional radio operation.

This led to the initial question:

For a fixed source-to-destination distance, is there an energy-optimal number of communication hops?

A second question followed from the same observation:

Does increasing the number of hops always improve energy efficiency, or is there a point after which the energy consumed by additional nodes becomes greater than the energy saved by shorter links?

These questions motivated the analytical model presented in this work. The initial goal was not to develop a complete routing protocol. Instead, the objective was to isolate the energy trade-off associated with hop count and establish a simple model that could be understood, derived mathematically, and tested computationally before introducing more realistic network and environmental effects.

A. Initial Reasoning

The first part of the reasoning considered only the effect of transmission distance. Under a free-space propagation assumption, the required transmit power increases approximately with the square of link distance,

$$P_t \propto d^2.$$

If a total source-to-destination distance D is divided into H equal hops, the distance of each hop is

$$d = \frac{D}{H}.$$

Therefore, the required transmit power for each hop scales as

$$P_t(H) \propto \frac{D^2}{H^2}.$$

At this point, considering RF transmit power alone would suggest that increasing the number of hops should continuously reduce the transmission cost. However, this result does not represent the complete energy consumption of a multi-hop system. Every additional hop requires a receiving operation and a transmitting operation at an intermediate node.

This led to an important modification of the model: circuit energy had to be included in addition to the RF transmission energy. With transmitter and receiver circuit power and finite power-amplifier efficiency included, the model contains two competing effects. The propagation-related energy decreases with increasing hop count, while circuit energy increases with the number of hops.

The resulting energy relationship can be written in the form

$$E(H) = \frac{A}{H} + BH.$$

This form provides a simple way to study the trade-off between using fewer long links and using more short links. It also makes it possible to calculate both a continuous stationary point and the physically meaningful integer hop count.

The remainder of this paper develops this model, verifies its basic physical scaling, and evaluates its behavior through a reproducible computational experiment suite.

II. SYSTEM MODEL

The V1 model considers a linear communication path between a source and destination with a fixed total separation D . The path is divided into H equal-length wireless hops. All nodes are assumed to use identical radio parameters.

For total distance D and H equal hops,

$$d = \frac{D}{H}.$$

The V1 propagation model assumes ideal free-space conditions. Friis transmission gives the required transmit power as

$$P_t(d) = \frac{P_{r,\min}}{G_t G_r} \left(\frac{4\pi d}{\lambda} \right)^2, \quad \lambda = \frac{c}{f},$$

where $P_{r,\min}$ is the minimum received power required by the receiver, G_t and G_r are the transmitter and receiver antenna gains, f is the carrier frequency, and c is the speed of light.

Substituting the equal-hop distance gives

$$P_t(H) = \frac{P_{r,\min}}{G_t G_r} \left(\frac{4\pi D}{H\lambda} \right)^2.$$

Thus, under the free-space assumption,

$$P_t(H) \propto H^{-2}.$$

The model uses a fixed receiver sensitivity and does not include fading, shadowing, interference, path-loss variation, atmospheric attenuation, or packet loss. These effects are deliberately excluded from V1 so that the relationship between hop count and energy can first be studied under controlled conditions.

A maximum transmitter power $P_{\max}$ is also included. A candidate hop count is considered feasible only when

$$P_t(H) \leq P_{\max}.$$

III. ENERGY FORMULATION

For a packet containing B bits transmitted at data rate R , the packet transmission time is

$$t_p = \frac{B}{R}.$$

The RF transmit power calculated from the propagation model is not the same as the electrical power drawn by the

transmitter. With a power-amplifier efficiency of η_{PA} , the power-amplifier contribution is

$$P_{PA} = \frac{P_t(H)}{\eta_{PA}}.$$

For H equal hops, the total PA energy required for one packet is therefore

$$E_{PA} = H \frac{P_t(H)}{\eta_{PA}} t_p.$$

Each hop also requires transmitter and receiver circuit operation. If the corresponding circuit powers are $P_{C,TX}$ and $P_{C,RX}$, the circuit-energy contribution is

$$E_C = H(P_{C,TX} + P_{C,RX})t_p.$$

The total packet energy is consequently

$$E(H) = H \left[\frac{P_t(H)}{\eta_{PA}} + P_{C,TX} + P_{C,RX} \right] \frac{B}{R}.$$

Since $P_t(H)$ is proportional to $1/H^2$, multiplying it by the number of hops gives an RF-related energy term proportional to $1/H$. The circuit term, on the other hand, is proportional to H . The total energy can therefore be written as

$$E(H) = \frac{A}{H} + BH$$

where A contains the propagation and PA-related parameters and B represents the per-hop circuit-energy contribution.

This form is useful because it makes the basic trade-off visible without hiding it inside the simulation code. The first term decreases as more hops are introduced, while the second term increases linearly. Therefore, the model does not assume that more hops are always better.

For the V1 model, packet size and data rate appear through the factor B/R . They affect the absolute energy required for packet transmission, while the hop-count behavior is primarily determined by the relative propagation and circuit-energy terms.

IV. OPTIMIZATION

Treating the hop count temporarily as a continuous variable gives

$$E(H) = \frac{A}{H} + BH.$$

Differentiating with respect to H ,

$$\frac{dE}{dH} = -\frac{A}{H^2} + B.$$

Setting the derivative to zero gives the continuous stationary point

$$H_0 = \sqrt{\frac{A}{B}}.$$

This result is useful for understanding the model, but a real communication path cannot contain a fractional number

of hops. The final optimization is therefore performed over positive integers.

In addition, a hop count may be physically infeasible if the required transmit power exceeds the transmitter's maximum output power. The implemented optimization is therefore

$$H^* = \arg \min_{\substack{H \in \mathbb{Z}^+ \\ P_t(H) \leq P_{\max}}} E(H).$$

The corresponding continuous preferred hop distance can be obtained by substituting $H = D/d$ into the energy expression. Defining

$$K = \frac{P_{r,\min}}{G_t G_r} \left(\frac{4\pi}{\lambda} \right)^2,$$

the preferred hop distance is

$$d_0 = \sqrt{\frac{\eta_{PA}(P_{C,TX} + P_{C,RX})}{K}}.$$

The quantity d_0 should not be interpreted as a universal optimum communication range. It is the characteristic hop distance predicted by the assumptions and radio parameters of the V1 model.

V. EXPERIMENTAL METHODOLOGY

The baseline uses 500 m, 2.4 GHz, -90 dBm receiver sensitivity, 10 mW TX and RX circuit power, 35% PA efficiency, a 100-byte packet, 1 Mbps data rate, and a 100-mW maximum TX power. Integer hop counts from 1 to 100 were evaluated.

Nine parameter sweeps were conducted: distance, RX circuit power, TX circuit power, frequency, PA efficiency, receiver sensitivity, packet size, data rate, and maximum TX power. The archived dataset contains 53 scenario summaries and 5,300 detailed hop-level evaluations.

A. Reproducibility and Experimental Record

The experiments were run as a separate V1 experiment suite rather than calculating individual values manually while preparing the paper. Each scenario evaluated the permitted integer hop counts and recorded the relevant input parameters, required transmit power, energy components, total energy, latency, feasibility, and selected optimum.

The numerical results were exported after the experiment run and retained as the V1 numerical record. This was done so that the values reported in the paper remain tied to a fixed experiment rather than depending on repeated simulation runs during writing. The archived results contain both scenario-level summaries and hop-level evaluations.

VI. VALIDATION

The free-space scaling test confirmed that doubling distance produces a fourfold RF transmit-power requirement, consistent with $P_t \propto d^2$. Energy decomposition also satisfied $E_{total} = E_{PA} + E_C$ to numerical precision.

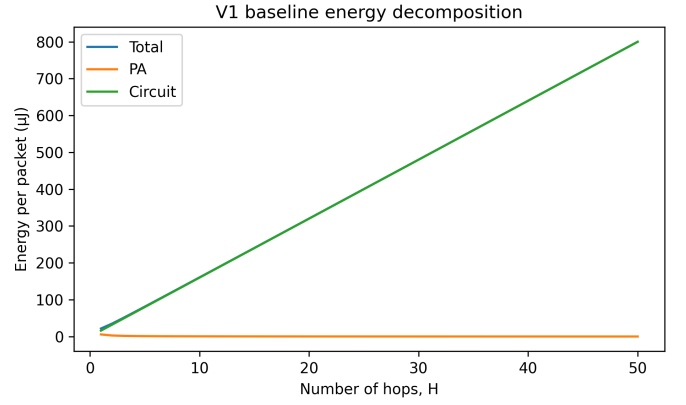


Fig. 1. Baseline energy decomposition. PA energy decreases with hop count while circuit energy increases.

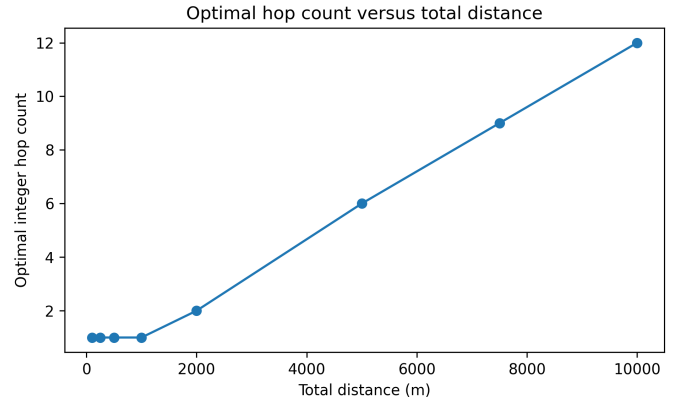


Fig. 2. Optimal integer hop count versus total distance.

VII. RESULTS

A. Baseline

For the baseline, $H_0 \approx 0.6012$ and the constrained integer optimum is

$$H^* = 1.$$

The required RF transmit power is approximately 4.031 dBm (2.53 mW), and the minimum packet energy is approximately 21.78 μ J.

B. Distance

The distance sweep shows the clearest topology transition: one-hop operation is optimal at short distances, while increasingly many hops become favorable at longer distances.

At 5 km, 7.5 km, and 10 km, the selected integer hop distance is approximately 833 m. This is consistent with the characteristic continuous hop distance implied by the analytical model.

C. Frequency

The frequency sweep shows that increasing frequency can shift the topology toward multi-hop operation because free-space required transmit power scales with f^2 .

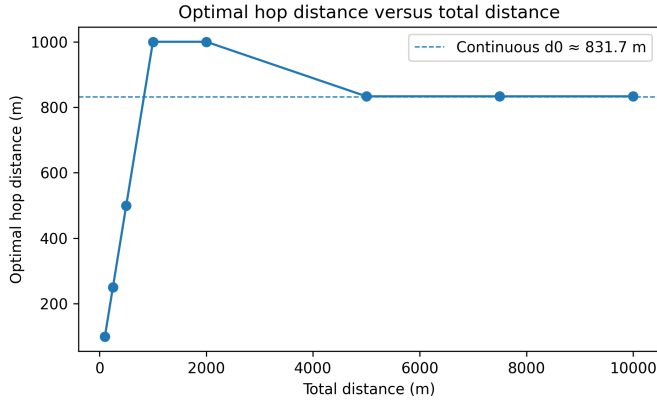


Fig. 3. Selected hop distance versus total distance.

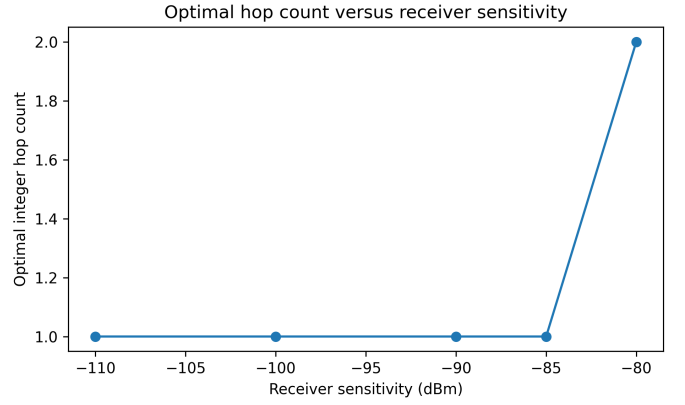


Fig. 5. Optimal hop count versus receiver sensitivity.

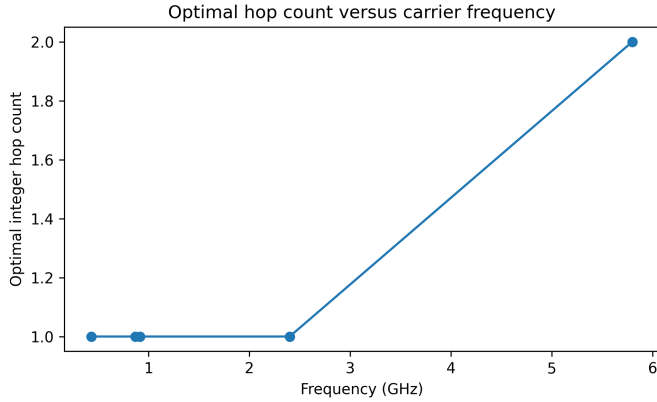


Fig. 4. Optimal hop count versus carrier frequency.

D. Receiver sensitivity

Increasing the required received power similarly increases the cost of long links and can cause a transition toward multiple hops.

Circuit-power and PA-efficiency sweeps changed energy magnitude but did not change the baseline optimal topology over the tested ranges. Packet size and data rate scale energy through B/R and did not produce a topology transition.

The maximum-TX-power sweep did not activate its constraint for the 500-m baseline because the smallest tested limit exceeded the direct-link RF requirement.

VIII. DISCUSSION

The main result of V1 is that hop count represents a trade-off rather than a quantity that should simply be maximized. The model gives

$$E_{PA} \propto H^{-1}, \quad E_C \propto H.$$

As additional hops are introduced, the propagation-related contribution decreases, but the circuit contribution increases approximately linearly. The energy optimum therefore depends on the relative size of these two effects.

This explains the behavior observed in the distance sweep. At relatively short distances, the energy required for a direct link is small enough that adding relay nodes does not justify their additional circuit consumption. At longer distances, the propagation cost of a direct link becomes more important, and additional hops can become energetically favorable.

The results also help answer the question that motivated the work. Increasing the number of hops is not inherently an energy-saving strategy. Instead, there is a balance between shorter transmission distances and the energy required to operate the additional nodes.

For sufficiently large source-to-destination distances, the optimizer can increase the number of hops while keeping the individual hop distance within a similar characteristic range. This behavior follows from the analytical form of the model and is visible in the distance results.

The model should not, however, be interpreted as a complete routing algorithm for a real mesh network. A real routing decision also depends on connectivity, packet loss, interference, latency requirements, node availability, queueing, and other network-level factors. V1 isolates only one part of that larger problem: the energy cost associated with hop count under the specified physical assumptions.

This separation is intentional. By establishing the idealized case first, later models can introduce additional effects and compare their results against the same baseline.

IX. LIMITATIONS AND FUTURE WORK

V1 uses free-space propagation, equal hop lengths, identical radios, fixed circuit power, fixed PA efficiency, and a serial packet-latency abstraction. It excludes fading, shadowing, interference, retransmissions, packet loss, MAC overhead, queueing, mobility, heterogeneous hardware, and environmental effects.

Future work will introduce environmental variables such as temperature, humidity, pressure, altitude, and visibility, while preserving V1 as the controlled baseline.

X. CONCLUSION

This work started from a practical question encountered while considering a multi-hop mesh communication system: for a fixed source-to-destination distance, how many hops should be used to transfer a packet efficiently?

The initial reasoning showed that shorter links require less RF transmit power under free-space propagation. However, considering transmission power alone would incorrectly favor continuously increasing the number of hops. Including transmitter and receiver circuit energy revealed the opposing effect of additional relay nodes.

The resulting analytical model has the form

$$E(H) = \frac{A}{H} + BH,$$

where the first term represents the decreasing propagation-related contribution and the second term represents the increasing per-hop circuit contribution. The model provides both a continuous stationary point and a constrained integer optimization problem.

The V1 computational experiments confirmed the expected free-space distance scaling and showed that the preferred topology depends on distance and radio parameters. For the 500-m baseline, the integer optimum was one hop with approximately $21.78 \mu\text{J}$ of energy per packet.

The purpose of V1 is therefore not to claim a universal optimal hop count. Instead, it provides a simple and reproducible reference model from which the effect of additional real-world factors can be studied.

Future work will retain V1 as the controlled baseline while extending the model toward environmental and channel-dependent conditions, including variables such as temperature, humidity, pressure, altitude, and visibility. This will allow the effect of these additional conditions to be studied relative to a clearly defined ideal case.

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